

Designing and Evaluating Explanations for a Predictive Health Dashboard: A User-Centred Case Study

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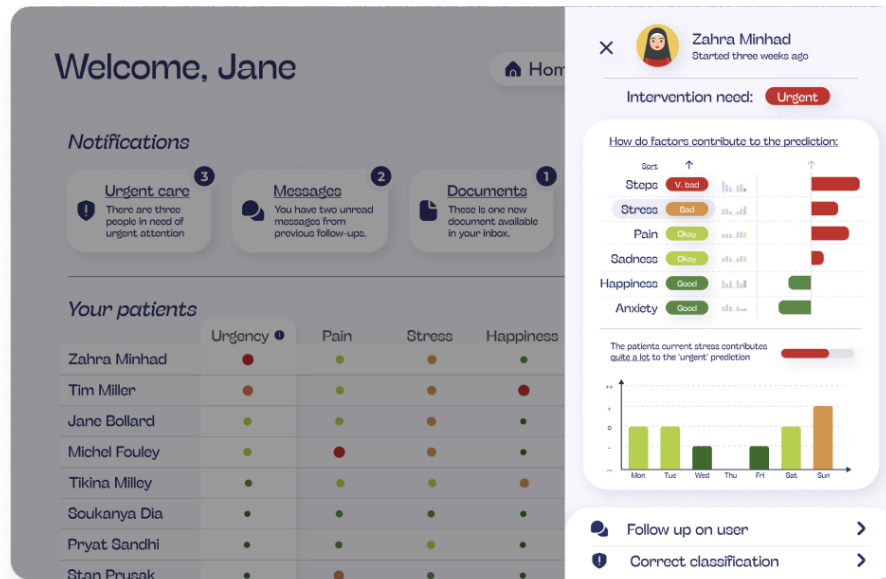


Figure 1: Health dashboard depicting predictions and explanation

ABSTRACT

As predictive health technologies become increasingly prevalent, the need for effective explanations that aid health experts and practitioners in understanding the underlying factors driving predictions is paramount. While many different explanation methods have been elaborated, recent research suggests that these explanations are often too complex for AI novices. In addition, little work has been done to evaluate whether the proposed methods indeed enhance human interpretability. In this study, we develop and evaluate explanations tailored to the needs of health experts, following an iterative user-centred design process to ensure understanding and usefulness of our explanation designs. Our findings underscore the importance of data-centric explanations, prioritising an emphasis on the underlying data rather than solely focusing on the

model's internal workings. Additionally, explanations should not only highlight points of congestion or areas for improvement but also emphasise positive aspects to promote a holistic understanding of the predictive health dashboard.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI; User studies; Empirical studies in visualization; Visualization design and evaluation methods.**

KEYWORDS

XAI, health dashboard, health explanations, domain experts, feature importance explanation, counterfactual explanations

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3637140>

ACM Reference Format:

Maxwell Szymanski, Vero Vanden Abeele, and Katrien Verbert. 2024. Designing and Evaluating Explanations for a Predictive Health Dashboard: A User-Centred Case Study. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3613905.3637140>

1 INTRODUCTION

The rise of AI in medical dashboards has brought numerous benefits to the healthcare industry by enabling faster, more accurate, and more efficient data analysis, leading to improved decision-making by healthcare professionals [17, 18]. They have the potential to enhance workflow and improve patient care by providing real-time predictions of adverse prognosis [17]. Additionally, AI offers great potential to support personalised healthcare interventions tailored to individuals or specific sub-groups of populations [11].

However, there are limitations when it comes to the understandability of AI-driven medical dashboards. The complexity of AI algorithms and the lack of transparency in their decision-making processes can make it challenging for healthcare professionals to fully comprehend and trust the outputs of these systems [10]. Explainable AI (XAI) tries to address this limitation by making AI transparent, enhancing understandability and enabling informed decisions. While XAI has shown promising results within the health domain, many existing XAI methods are tailored towards AI experts [8], causing a significant lack of understanding from AI novices. Furthermore, work of Szymanski et al. [15] has shown that AI novices can be unaware of their lack of understanding, potentially leading to incorrectly informed decisions. In addition, despite the rich set of XAI algorithms and off-the-shelf toolkits for AI developers, successful examples of XAI are still relatively scarce in real-world AI applications [7]. Prominent work in this area of Liao and Varsney [7] highlights the disconnect between technical XAI approaches and supporting users' end goals in usage contexts and the need for user-centered design efforts that fill this gap. This work is increasingly researched under the umbrella of human-centred XAI, where tools are designed together with stakeholders, ensuring broader understanding and participation [7]. These human-centred XAI approaches are gaining increased research attention to enable the adoption of AI in real-life settings [9].

In this paper, we address the identified gap between technical XAI approaches and user needs in the context of an mHealth application that we developed together with IDEWE, the largest occupational health service provider in Flanders. During an extensive three year project, we developed an mHealth application that is able to coach and inform users suffering from chronic musculoskeletal pain. The main function of the application is to monitor and educate users through a variation of information, questionnaires and interactive exercises, by guiding them through content modules regarding various topics, such as activity management, pain-education, stress and emotions. Users receive daily short questionnaires regarding their pain and emotional state, as well as a daily coaching module regarding one of the topics. This data is then used to predict whether employees may be at risk of long term absenteeism. To enable the adoption of these predictions in real-life settings by the health experts of IDEWE, we designed and evaluated explanations tailored to their needs, following an iterative user-centred design process to ensure understanding and usefulness of our explanation designs (as described in Figure 3, the studies have been ethically approved by the Ethical Committee of UZ Leuven with application number S65610.).

The research contributions of this work are three-fold:

- First, we present the design of novel simplified explanation methods that can be used by AI novices to support understandability of model outcomes. We present improved counterfactual and feature importance explanation designs that are less complex for health experts.
- Second, we present the results of an iterative user-centred design process of a transparent health dashboard that supports health experts in their intervention efforts. The health dashboard gives experts a comprehensive overview of users and identified individuals who may require intervention. To provide context and facilitate understanding for health experts, the dashboard uses the novel simplified explanation methods.
- Third, we present the results of an evaluation of the effectiveness and user perception of different explanation methods as well as design implications that can be used by researchers and practitioners in the domain.

2 PHASE 1: DEVELOPING A HEALTH APP AND DASHBOARD

In this section, we present an overview of our developed health application designed for monitoring employees with chronic musculoskeletal pain and the development of a health dashboard to support health experts in identifying employees who may require intervention. Furthermore, we explore the concept of explanations as a means to facilitate informed decision-making for health experts in their interactions with employees.

2.1 Mobile Health App

Before we describe our health dashboard, we briefly explain the content and functionality of the health application we developed to support employees with chronic musculoskeletal pain. The app (Figure 2) consists of a coaching component, consisting of several modules that give the user information regarding chronic pain and how to manage it, and a monitoring component, to follow up on a user's progress through daily brief questionnaires as well as more elaborate 6-weekly ones. The app is made available on both the App Store and Play Store for convenient access. However, during the initial development phase, registration was restricted to invite-only, ensuring participation was limited to those specifically contacted. We invited 322 participants to use the application for at least half a year in order to gather feedback on the design and content of the application, as well as gather pseudonymised daily questionnaire data to be used for designing the explanations. They have all been given a FitBit Inspire in order to capture activity data as well.

Coaching component. The coaching component of the system encompasses six distinct modules designed to address topics relevant to understanding and managing chronic musculoskeletal pain. These modules include pain education, stress and resilience, social contexts, thoughts and emotions, movement, as well as work and physical activity. Each module is structured to provide users with information conveyed through textual content, videos, interactive questions, and exercises.

Furthermore, the coaching component offers a pain logbook feature that enables users to record episodes of pain flare-ups using a conversational agent. This agent engages the user in a dialogue,

inquiring about their thoughts, emotions, and responses associated with the experienced pain. Subsequently, the agent generates personalised recommendations aimed to assist the user in coping with similar episodes in the future. These recommendations also direct the user towards relevant sub-modules within the corresponding modules. For instance, if the user expresses feelings of tension, the agent may guide them to explore the mindfulness sub-module within the thoughts and emotions module.

Monitoring component. In addition to the coaching component, it is important to monitor the progress of users in their chronic pain journey. To achieve this, we use several assessment measures to obtain an overview of the user's status. These assessments include daily questionnaires, measuring pain intensity (through a Visual Analogue Scale, VAS, ranging from 0 to 100) [4], as well as assessments of positive and negative emotions using the Positive and Negative Affect Schedule (PANAS) [19]. Furthermore, users are asked to report their perceived load and capacity using the VAS, as well as whether or not they worked that day. Physical activity was tracked through the FitBit Inspire all 322 participants received, which records the number of steps taken and at which time. All measured parameters were refined in previous work with health experts through iterative design where we assessed their relevance for predicting long-term absenteeism [6]. By incorporating these assessments, we aim to gather extensive data that enables a thorough evaluation of the user's progress throughout the chronic pain management program.

2.2 Design Rationale for the Health Dashboard and Collected Data

The integration of coaching modules and components, such as the pain logbook, within the app allows for guidance and support to users. However, it is important to recognise that there are instances where intervention by a healthcare expert is necessary, particularly when users do not exhibit improvements or experience a deterioration in pain symptoms or other measurements, and in practice is often done through a dashboard [1]. Interventions with app users can be broadly interpreted to include personalised messages, reminders to revisit specific modules (e.g., mindfulness in cases of prolonged high-stress levels), or invitations for in-person consultations. In order to assist the expert in identifying users who may require intervention, we extend the dashboard with a predictive parameter "urgency". This parameter, derived from the user's daily questionnaire data as well as their FitBit data, will serve as an indicator of the level of urgency for intervention. As predictive systems can be subject to errors or calculations that may not be in line with what a health expert would expect, previous research has emphasised the significance of transparency in such systems, particularly through the application of explainable AI [2].

In short, our dashboard offers an overview of the monitored parameters in the in-app daily questionnaires, including daily perceived pain, stress, anxiety, and the amount of steps taken, enabling practitioners to gain a comprehensive understanding of patients' conditions. To populate this data, we use real-world data collected from the HealthEmpower smartphone app during a 6-month study involving 322 participants, where they were asked to interact with the coaching modules and fill in the daily questionnaires as often as

possible. Moreover, the dashboard shows an indicator of whether or not a particular user could be deemed urgent for intervention.

2.3 Explainable AI Methods for Health Dashboards

To make the predictions in our health dashboard transparent, we rely on guidelines from previous research that highlight the suitability of the following two prominent types of explanations for health experts: feature attributions and counterfactuals [18] (Figure 4 left). Feature attributions provide insights into the contribution of each parameter towards the prediction, whether it is a positive or negative influence. For instance, while a low number of steps may initially appear alarming, its contribution may be slightly lower compared to a high pain score, as the system places greater emphasis on the latter's contribution. Through seeing these feature attributions, health experts can gain insights into how the predictive system reasons about each individual parameter and make guided decisions regarding whether to intervene or not.

Additionally, counterfactual explanations are used to indicate the potential changes in parameters and how they should change in order to reduce the urgency for intervention. Counterfactual reasoning often aligns with human reasoning processes, thereby facilitating interpretation and understanding [3]. Moreover, counterfactual explanations can offer direct suggestions to health experts regarding the specific parameters that require attention and how they should be addressed (i.e. the dashboard would suggest increasing average steps by a 1000, or decrease the perceived workload by 10%, in order to decrease the need for urgency). These suggestions can serve as an initial indication for designing a targeted intervention.

3 PHASE 2 - IMPROVING CF AND FI EXPLANATION DESIGNS

To assess our initial explanation designs, we conducted a co-design session with 6 healthcare experts and 2 researchers in the clinical field. Using the results from this co-design session, we synthesised some takeaways and use these to adapt our designs for the following iteration.

3.1 Methodology

To evaluate the effectiveness of Counterfactual (CF) and Feature Importance (FI) explanations in the health dashboard, we organised a co-design session acting as a brainstorming session. Six occupational therapists and two healthcare researchers, who were actively involved in developing the HealthEmpower smartphone app, participated in the workshop. Our primary objectives for this workshop were to evaluate the understandability and practicality of the explanations provided and to gather insights that would guide refinements in the dashboard's next design iteration. Considering the COVID-19 regulations, the workshop was conducted online, ensuring the safety of all participants, and lasted 60 minutes. Participants had access to the high-fidelity Figma prototype¹ and were asked to freely interact with it whilst having an open discussion.

¹Figma prototype: <https://wms.cs.kuleuven.be/cs/onderzoek/augment/demos>

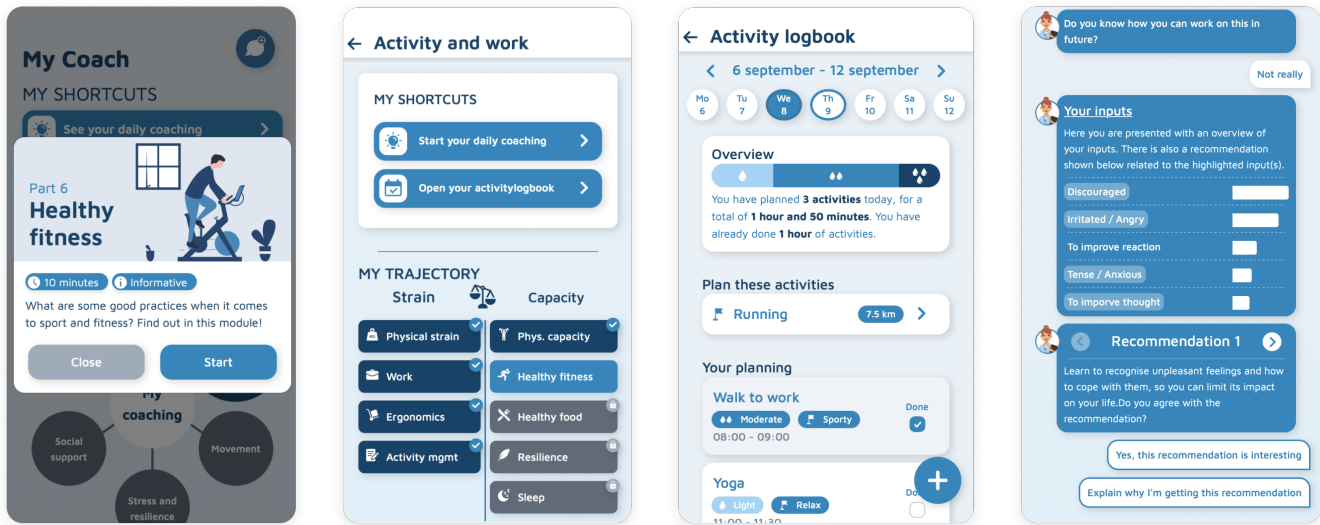


Figure 2: The mobile HealthEmpower app, displaying the pain logbook and coaching components

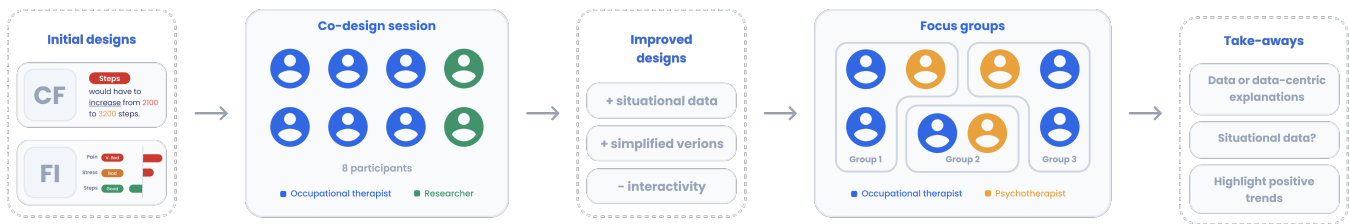


Figure 3: Flow of the case study, including design phase as well as a co-design session and focus group study

Session notes were taken, and an online collaborative tool Miro² was used to facilitate information sharing and collaboration during the workshop.

3.2 Design improvements

The CF and FI explanations received an overall positive reception from participants during the workshop. However, several improvements and adaptations were suggested from the evaluation, which we summarise as follows:

- **Low need for model improvement:** Users expressed a slight concern about the learning curve of the explanations. Adding interaction to facilitate model improvement was considered to be too difficult to understand. Participants mentioned that instead of spending time fine-tuning model features, they would rather delve into additional information to identify whether or not an intervention is actually needed.
- **Simplifying Feature Importance Explanation Design:** Most participants expressed that while they did not necessarily require precise feature attribution values, they still desired a general ordering of feature attributions to understand which parameters contribute the most. Simplifying the design of feature importance explanations to provide

a relative ranking of feature contributions was considered sufficient according to the health experts.

- **Introducing Data Previews:** Two participants emphasised the need for quick data insights at a glance. One suggestion was to incorporate a minimalistic line graph next to each parameter in the Feature Importance overview, indicating the trend (upward, downward, constant, or zig-zagged). Another participant recommended displaying a simplified bar chart next to each parameter, providing a visual representation of a weeks-worth of data. These data previews were seen as beneficial to quickly grasp the status and trends of monitored parameters.

4 PHASE 3: ASSESSING CF AND FI EXPLANATION DESIGNS

To evaluate the effectiveness and user perception of Counterfactual (CF) and Feature Importance (FI) explanations, we conducted a study involving three focus groups comprising occupational therapists and psychotherapists. The study aimed to assess their perspectives on the need for a predictive dashboard and their preferences regarding CF and FI explanations.

²Online collaborative tool: <https://miro.com/>

4.1 Methodology

We designed three focus groups and recruited a total of eight occupational therapists and psychotherapists to participate in the study. An overview of participants is given in Table 1. The primary objective was to gauge their requirements for a predictive dashboard and their perception of CF and FI explanations. The focus groups began by engaging in a discussion centred around the concept of FI explanations. Participants were introduced to the original FI explanation design and provided with an opportunity to explore and comprehend its functionality and relevance within the context of the healthcare domain. This was followed by a discussion of the simplified FI explanations and counterfactual explanations. To conclude the study, participants were asked to express their preference between the two explanation modalities (FI and CF) and justify their choice. This allowed for a deeper understanding of the factors influencing their preferences and their specific needs in terms of explanation functionalities.

4.2 Results

We will now discuss the codes and themes extracted through an iterative, inductive thematic analysis. An overview of extracted themes is given in Table 2.

Model improvement & insight. The first theme relates to model insights and improvement. By using explanations to make the underlying and the already included parameters (i.e. steps, stress) transparent, participants were able to give **suggestions regarding the included parameters**. For example, P_4 and P_7 suggested sleep as an important parameter to be included in the model (be it self-monitored or through a wearable), and P_2 deems workload to be an important factor for monitoring chronic pain patients. Some participants also expressed concerns with **including subjective parameters**, such as their emotional state, in such a model, with P_7 explaining it as: “I had pain for a while, and some days you feel better than others, not every day can be a ten out of ten. And then to emphasise that you should feel good... It’s okay not to sometimes”. This issue was more apparent in the counterfactual setting due to the way the parameters are presented, with P_8 mentioning: “I feel a lot of resistance within myself when I see this suggestion that sadness needs to go from quite high to moderate. With steps or stress, you do have clear interventions to tackle that problem, whereas it’s not so clear-cut with happiness or sadness”.

Intervention needs. A second theme relates to what a model needs to provide in order to support an effective intervention. P_4 and P_6 appreciated how counterfactuals were able to give **concrete suggestions for interventions** and would use those as initial starting points. P_6 describes this as: “I like the fact that it’s actually formulated as a goal, which makes it easier for me to know what my next [intervention] steps are”. The explanations were also able to **show areas of improvement** at a glance (FI more so than CF), with P_2 mentioning: “I don’t think I can already give a concrete action plan just by looking at this, but I also don’t think that’s the point. What I do have is a clear overview of the points of contention with these inquired parameters”. However, most participants expressed the need for **more contextual information** to provide effective interventions. P_2 mentions side information such as the sector they work in (e.g.

desk job) to give appropriate feedback regarding the number of steps taken, as well as contextual information that the user should log: “If I see that stress is high, I would like to know whether that is work-related or not. Otherwise, it’s hard to [appropriately intervene]”. P_2 adds to this by saying: “With the week-overview, I see that the steps are low, but is it logical to start an intervention or not? Because a desk worker will probably not reach their step goal”. P_4 furthermore expresses the need for additional textual input: “When we are planning interventions in a reload-trajectory, we ask our patients to i.e. describe their activities, or sleep in the last few weeks in a textual way to get a proper storyline. [...] And bad parameters don’t always correlate to a need for intervention as there might be a lot of progression going on, but this model is assuming bad parameters correlate with an urgent need for intervention. So that is why I need this extra context”.

Data-centric information. To facilitate more informed interventions, participants expressed a strong need for either more data-centric explanations or more raw data/information regarding trends. Visualising trends would help experts make more informed decisions, with P_7 explaining: “It’s like on the smartwatch, where you’re getting overviews, and also see trends, that is interesting to see”. Explanations rooted in data could help explain these trends according to some participants, with P_6 mentioning: “I would like an explanation regarding the trend in the data, is it just one bad day, multiple days, or a fluctuating pattern? I feel like that information is more useful than [the feature importances]”. P_7 adds to this by saying that explanations could focus on summarising patterns and evolutions in a patient’s data. Finally, showing averages, from either the patient themselves or all patients, was also suggested as a way to indicate where the current patient is situated.

4.3 Preference for explanation type

During the focus groups, participants were evenly divided in their preference for counterfactual (CF) explanations and feature importances (FI). Those who favoured CF explanations appreciated their ability to provide concrete suggestions, which could be used as a foundation to plan interventions. Conversely, the other half disliked CF due to its “suggestive nature”, gravitating instead towards FI explanations. Another critique of CF was its tendency to highlight only areas for improvement, inherently neglecting positive attributes. This was seen as a shortcoming, especially for long-term interventions, resulting in a preference for FI explanations. When given a choice between the standard FI and a simplified version, 6 out of 8 participants preferred the latter. Their reasons were twofold: (1) they found the actual values of importances in the standard FI either too intricate, or (2) believed they didn’t provide any practical insight for planning interventions with patients.

5 DESIGN IMPLICATIONS

Given the results from our thematic analysis, we will now discuss some design considerations relevant for designing health explanations for health experts. From the thematic analysis, we are able to extract five general guidelines on how to design effective explanations for healthcare experts in long-term interventions.



Figure 4: Overview of initial explanation designs, as well as improvements made to the explanations in magenta by (1) introducing data previews and (2) simplifying feature importances

Table 1: Overview of participants during the final focus groups

ID	Occupation	Age	Gender	Preference
P ₁	Prevention advisor ergonomics	51-60	F	FI
P ₂	Prevention advisor ergonomics	31-40	F	FI
P ₃	Prevention advisor psychosocial wellbeing	31-40	M	CF
P ₄	Prevention advisor stress & burn-out	41-50	M	CF
P ₅	Ergonomist, clinical care	31-40	M	CF
P ₆	Prevention advisor ergonomics	31-40	F	CF
P ₇	Prevention advisor psychosocial wellbeing	31-40	F	FI
P ₈	Occupational psychotherapist	31-40	F	FI

Table 2: Overview of codes and themes from the thematic analysis (FI - Feature Importance, CF - Counterfactuals, BT - Both)

	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈
Model improvement & insight								
Disagrees with outcome	FI	CF		FI				
Disagrees with included measurements	FI	CF						
Problem with subjective measurements							BT	BT
Intervention needs								
Suggests concrete action plans				CF		CF		
Shows area of improvement	FI		BT	FI		FI		
Needs more contextual information	FI		FI		FI	BT		BT
Data-centric information								
Needs information regarding trends				BT	FI	BT	BT	
Needs raw or historical data	FI			FI		FI		

Relative Feature Attributions Suffice. In our initial workshop, health-care practitioners indicated minimal necessity for specific values of feature attributions pertaining to the parameters studied. Consequently, we introduced a design emphasising relative feature importance without communicating actual importance values. In

subsequent focus groups, 6 out of 8 participants favoured this simplified design over detailed feature importance explanations, deeming specific attribution values either complex to understand or superfluous for intervention planning.

Focus on Positive Parameters. Our participants stressed the importance of not solely focusing on areas of concern but also acknowledging positive improvements observed in their patients. Counterfactual explanations, by their inherent nature, primarily addressed negative parameters and suggested interventions for improvement. However, participants underscored the need to incorporate positive aspects and interventions into the explanations, allowing for a more comprehensive assessment of patient progress and enabling practitioners to underline and reinforce positive outcomes. This differs from other studies with regards to XAI in healthcare that focus more on immediate interventions (i.e. glucose monitoring [1], or acute myocardial infarction [18]), where design guidelines underline the highlighting of points of contention. This is most likely due to the nature of the intervention - the aforementioned examples focus more on immediate interventions, whereas our study focused on XAI for long-term health monitoring, where the rehabilitation process is often non-linear. Thus given the feedback from our focus group, we conclude that in long-term monitoring, positive progress is equally as important as the points of contention, and should therefore be included in the design of explanations.

Need for Data-centric Explanations. Health-care practitioners expressed a strong desire for more raw data and visualisations, emphasising the need for comprehensive information beyond model explanations. While they appreciated the explanations, they viewed them as initial suggestions rather than the sole basis for decision-making. Participants believed that presenting the raw data, without extensive explanations, would be sufficient as they could leverage their domain expertise to interpret and make informed decisions. The data-previews that were incorporated after the initial workshop were appreciated, however, further steps are needed. Augmenting raw data visualisations with trend lines or annotations was suggested to enhance data interpretation and facilitate a deeper understanding. Previous work of Holzinger et al. mentions that explanations for health experts should be based on interpretable and meaningful data, enabling them to understand the underlying factors and trends influencing health predictions [5]; however, our participants mention that additional steps can be taken beyond explanations by also showing raw data, accompanied by visualisations and trend-lines, to make more informed decisions.

Appropriate use of Model-centric Explanations for Domain Experts. We find themes that allude to certain shortcomings in the predictive model that our participants were able to identify through FI and CF explanations. Work in user-centred XAI mentions that model debugging, verification and improvement are tasks attributed to AI practitioners, whereas domain experts mainly use XAI to understand the underlying inferences [12, 16]. Interestingly, we observed that domain experts found limited value in understanding the model through feature importance. However, through these model-centric explanations, participants were able to get an overview of which parameters are included in the model and - in the case of feature importances - how much they contribute to the outcome prediction. These insights enabled participants to propose additional measurements relevant for pain-related interventions, such as incorporating data on sleep and workload. This suggests that, while model-centric explanations had limited practical use, domain experts could leverage these explanations in an earlier stage - e.g. the model training

stage - to improve the model accordingly given their domain expertise.

Problems with Subjective Health Measurements. Health-care practitioners showed scepticism towards the inclusion of certain subjective health measures in the predictive model and their corresponding explanations. For instance, one participant highlighted the difficulty of quantifying anxiety, which is a layered and multifaceted construct. Consequently, they suggested that subjective measures, such as anxiety, should be considered as contextual information rather than influencing the predictions and being extensively explained. They preferred these parameters to be supplementary data, providing valuable context rather than directly impacting the model's predictions. Other studies in other domains, such as oncology and cardiology, have also highlighted problems of using self-reported data in AI models, as they need more validation compared to objective measures [13, 14]. This highlights the need to handle subjective or self-reported data with more care compared to objective measures, or as some experts in our study denoted, solely use them as contextual indicative data and not predictive data. However, relating to the previously mentioned need for contextual data, these should still be communicated through trend lines or raw data to support more informed decisions.

6 CONCLUSION

Our work focused on the development of a transparent health dashboard integrated with a mobile app aimed at supporting health experts in their intervention efforts. The health dashboard gave experts a comprehensive overview of users and identified individuals who may require intervention. To provide context and facilitate understanding for health experts, we designed explanations using an iterative and user-centred approach.

During the co-design sessions, experts emphasised the importance of situational data and requested simplified representations of health explanations. Taking these insights into account, we iterated over our explanation designs and conducted a follow-up focus group study with 8 health experts in total. The results showed that health experts heavily rely on data and expressed a preference for explanations that are more data-centric in conjunction with the existing model-centric methods. Additionally, they expressed their need for a representation that highlights positive trends in the data, not solely focusing on factors contributing to the urgency of intervention, which is inherent to e.g. counterfactual recommendations. Furthermore, our findings indicated that health-care practitioners displayed scepticism towards incorporating certain subjective health measures in the predictive model and their corresponding explanations. Specifically, subjective measures, such as anxiety, were suggested to be treated as contextual information rather than direct influencing predictions, providing valuable context without heavily impacting the model's predictions.

We would like to add that data in our study was collected from participants through the HealthEmpower mobile app. Since all of them were working at that time, the amount of participants actually needing urgent intervention was low, potentially skewing our predicted labels. Replicating these results with different datasets, and in different contexts, would prove useful for future work.

ACKNOWLEDGMENTS

We would like to thank all participants for their time and valuable insights. This research is part of the research projects Personal Health Empowerment (PHE) with project number HBC.2018.2012, financed by the Research Foundation Flanders (FWO) with project numbers G0A4923N and G067721N, as well as the C1 project with project number C14/21/072, and has been ethically approved by the Ethics Committee Research UZ / KU Leuven (EC Research) with application number S-65610.

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